
Consuming Senders from Coroutine-Native Code

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Audience:	LEWG
Reply-to:	Vinnie Falco vinnie.falco@gmail.com Steve Gerbino steve@gerbino.co

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Abstract

A coroutine type satisfying `IoAwaitable` ([P4003R0](#)^[3]) consumes `std::execution` senders with zero allocations, correct stop token propagation, and automatic executor dispatch-back. The bridge is one class template. The complete implementation is in Appendix A.

Revision History

R0: March 2026 (post-Croydon mailing)

- Initial version.
-

1. Disclosure

The authors developed and maintain [Capy](#)^[4] and [Corosio](#)^[6] and believe coroutine-native I/O is the correct foundation for networking in C++. The findings in this paper are structural and hold regardless of which library implements the coroutine-native layer. This paper is one of a suite of four that examines the relationship between compound I/O results and the sender three-channel model. The authors provide information, ask nothing, and serve at the pleasure of the chair. [Capy](#)^[4] is a coroutine primitives library. It contains no sockets, no timers, and no platform-specific I/O APIs. Networking is in [Corosio](#)^[6], which is not used in this paper. The bridge depends on [Capy](#)^[4] and [beman::execution](#)^[5], a community implementation of [std::execution](#) (P2300R10^[1]). Source code links refer to pinned commit [60bf781](#)^[4].

2. The Bridge

```
copy::task<int> compute(auto sched)
{
    int result = co_await copy::await_sender(
        ex::schedule(sched)
        | ex::then([] {
            std::cout
                << " sender running on thread "
                << std::this_thread::get_id()
                << "\n";
            return 42 * 42;
        }));

    std::cout
        << " coroutine resumed on thread "
        << std::this_thread::get_id() << "\n";

    co_return result;
}
```

`await_sender` returns a `sender_awaitable` that satisfies `IoAwaitable` (P4003R0^[3]). Any coroutine type whose promise propagates `io_env` through `await_suspend(h, io_env const*)` can use it. `copy::task` is one such type. It is not the only one.

The bridge is one class template. The complete implementation is in Appendix A. It was tested against `beman::execution` ^[5].

3. Demonstration

```
main thread: 32208
  sender running on thread 9560
  coroutine resumed on thread 34356
result: 1764
```

The sender executed on thread 9560, the `beman::execution::run_loop` thread. The coroutine resumed on thread 34356, the Capy thread pool. The value 1764 crossed the bridge. The bridge allocated zero bytes.

4. What the Bridge Does

The bridge consumes any `std::execution` sender. Scheduler hops work. Stop token propagation works.

`set_value` , `set_error` , and `set_stopped` are handled. The operation state is stored inline with zero allocations. Any sender pipeline - `when_all` , `then` , `let_value` , `on` - works through the bridge.

An `IoAwaitable` coroutine can suspend into a sender pipeline that schedules work on a GPU, a thread pool, or any other execution context. When the sender completes, the bridge dispatches the resumption back to the coroutine's originating executor (`env_->executor.post(cont_)`). The coroutine resumes in the correct context with the sender's result, regardless of where the sender executed. This is the interop path for coroutine-native I/O code that needs to offload compute to a GPU scheduler and resume on the I/O event loop.

The bridge does not use `execution::task` .

5. What the Bridge Does Not Require

Property	<code>execution::task</code>	Bridge
Routine I/O errors become exceptions	Yes	No
Type erasure on connect	Yes	No
<code>dispatch</code> adapter for compound results	Yes	No
Zero allocations in the bridge	No	Yes

`std::execution::task` is not necessary to consume senders.

6. The Narrowest Abstraction

`Capy`^[4] is a coroutine primitives library. It provides `task` , executors, stop tokens, frame allocators, and buffer sequences. It contains no sockets, no timers, and no platform-specific I/O APIs. Networking is in `Corosio`^[6], a separate library that depends on `Capy`^[4].

The sender bridge depends on `Capy`^[4] and `std::execution`. It does not depend on `Corosio`^[6]. It does not depend on any platform I/O API. A user who needs I/O adds `Corosio`^[6]. A user who needs sender interop adds the bridge. A user who needs neither does not pay for either. The coroutine type does not change between these configurations.

7. Acknowledgments

The authors thank Dietmar Kühl for `beman::execution`^[5] and for the channel-routing enumeration in `P2762R2`^[7], Michał Dominiak, Eric Niebler, and Lewis Baker for `std::execution`, Chris Kohlhoff for identifying the partial-success problem in `P2430R0`^[8], Kirk Shoop for the completion-token heuristic analysis in `P2471R1`^[9], Fabio Fracassi for `P3570R2`^[10], Ville Voutilainen and Jens Maurer for reflector discussion on dispatch patterns, Herb Sutter for identifying the need for tutorials and documentation, Mark Hoemmen for insights on `std::linalg` and the layered abstraction model, and Peter Dimov for the refined channel mapping.

References

1. `P2300R10` - “std::execution” (Michał Dominiak et al., 2024). <https://wg21.link/p2300r10>
2. `P3552R3` - “Add a Coroutine Task Type” (Dietmar Kühl, Maikel Nadolski, 2025). <https://wg21.link/p3552r3>
3. `P4003R0` - “IoAwaitable” (Vinnie Falco, 2026). <https://wg21.link/p4003r0>
4. `cppalliance/capy` - Coroutine primitives library. Commit f046646.
<https://github.com/cppalliance/capy/tree/f046646e63baf0cc3d6034bc35eec24694f5d16>
5. `bemanproject/execution` - Community implementation of `std::execution`.
<https://github.com/bemanproject/execution>
6. `cppalliance/corosio` - Coroutine-native networking library. <https://github.com/cppalliance/corosio>
7. `P2762R2` - “Sender/Receiver Interface For Networking” (Dietmar Kühl, 2023). <https://wg21.link/p2762r2>
8. `P2430R0` - “Partial success scenarios with P2300” (Chris Kohlhoff, 2021). <https://wg21.link/p2430r0>
9. `P2471R1` - “NetTS, ASIO and Sender Library Design Comparison” (Kirk Shoop, 2021).
<https://wg21.link/p2471r1>
10. `P3570R2` - “Optional variants in sender/receiver” (Fabio Fracassi, 2025). <https://wg21.link/p3570r2>

Appendix A. Bridge Implementation

```
#include <boost/capy/ex/io_env.hpp>

#include <beman/execution/execution.hpp>

#include <cassert>
#include <coroutine>
#include <cstring>
#include <exception>
#include <new>
#include <stop_token>
#include <tuple>
#include <type_traits>
#include <variant>

namespace boost::capy {

namespace detail {

struct stopped_t {};

struct bridge_env
{
    std::stop_token st_;

    auto query(
        beman::execution::get_stop_token_t const&)
        const noexcept
    {
        return st_;
    }
};

template<class Sender>
using sender_single_value_t =
    beman::execution::value_types_of_t<
        Sender,
        bridge_env,
        std::tuple,
        std::type_identity_t>;

template<class ValueTuple>
struct bridge_receiver
{

```

```

using receiver_concept =
    beman::execution::receiver_t;

std::variant<
    std::monostate,
    ValueTuple,
    std::exception_ptr,
    stopped_t>*      result_;
std::coroutine_handle<> cont_;
io_env const*      env_;

auto get_env() const noexcept -> bridge_env
{
    return {env_->stop_token};
}

template<class... Args>
void set_value(Args&&... args) && noexcept
{
    result_->template emplace<1>(
        std::forward<Args>(args)...);
    env_->executor.post(cont_);
}

template<class E>
void set_error(E&& e) && noexcept
{
    if constexpr (
        std::is_same_v<
            std::decay_t<E>,
            std::exception_ptr>)
        result_->template emplace<2>(
            std::forward<E>(e));
    else
        result_->template emplace<2>(
            std::make_exception_ptr(
                std::forward<E>(e)));
    env_->executor.post(cont_);
}

void set_stopped() && noexcept
{
    result_->template emplace<3>(
        stopped_t{});
    env_->executor.post(cont_);
}
};

```



```

} // namespace detail

template<class Sender>
struct sender_awaitable
{
    using value_tuple =
        detail::sender_single_value_t<Sender>;
    using receiver_type =
        detail::bridge_receiver<value_tuple>;
    using op_state_type = decltype(
        beman::execution::connect(
            std::declval<Sender>(),
            std::declval<receiver_type>()));

    Sender sndr_;

    std::variant<
        std::monostate,
        value_tuple,
        std::exception_ptr,
        detail::stopped_t> result_{};

    alignas(op_state_type)
    unsigned char op_buf_[sizeof(op_state_type)];
    bool op_constructed_ = false;

    explicit sender_awaitable(Sender sndr)
        : sndr_(std::move(sndr))
    {
    }

    sender_awaitable(sender_awaitable&& o)
        noexcept(
            std::is_nothrow_move_constructible_v<
                Sender>)
        : sndr_(std::move(o.sndr_))
    {
        assert(!o.op_constructed_);
    }

    sender_awaitable(
        sender_awaitable const&) = delete;
    sender_awaitable& operator=(
        sender_awaitable const&) = delete;
    sender_awaitable& operator=(
        sender_awaitable&&) = delete;

```

```

~sender_awaitable()
{
    if(op_constructed_)
        std::launder(
            reinterpret_cast<op_state_type*>(
                op_buf_))->~op_state_type();
}

bool await_ready() const noexcept
{
    return false;
}

std::coroutine_handle<>
await_suspend(
    std::coroutine_handle<> h,
    io_env const* env)
{
    ::new(op_buf_) op_state_type(
        beman::execution::connect(
            std::move(sndr_),
            receiver_type{
                &result_, h, env}));
    op_constructed_ = true;
    beman::execution::start(
        *std::launder(
            reinterpret_cast<
                op_state_type*>(
                    op_buf_)));
    return std::noop_coroutine();
}

auto await_resume()
{
    if(result_.index() == 2)
        std::rethrow_exception(
            std::get<2>(result_));
    if(result_.index() == 3)
        throw std::runtime_error(
            "sender completed with "
            "set_stopped");

    if constexpr (
        std::tuple_size_v<
            value_tuple> == 0)
        return;
}

```

```

        else if constexpr (
            std::tuple_size_v<
                value_tuple> == 1)
            return std::get<0>(<
                std::get<1>(<
                    std::move(result_))));
        else
            return std::get<1>(<
                std::move(result_));
    }
};

template<class Sender>
auto await_sender(Sender&& sndr)
{
    return sender_awaitable<
        std::decay_t<Sender>>(<
            std::forward<Sender>(sndr));
}

} // namespace boost::copy

```